

SYSTEMANALYSE & ENTWURF

Dozent: Gregor Tielsch
Kurs: 24SEA

IT'S DEVELOPER'S
(MIS)UNDERSTANDING,
NOT EXPERT
KNOWLEDGE THAT GETS
RELEASED IN
PRODUCTION

I'd like to
order a cake

Sure, what would you
like written on it?

Could you write "So
long, Alicia" in
purple?

Sure

And put stars
around it

No problem, I
took notes and
hand them over
to our cake
decorator





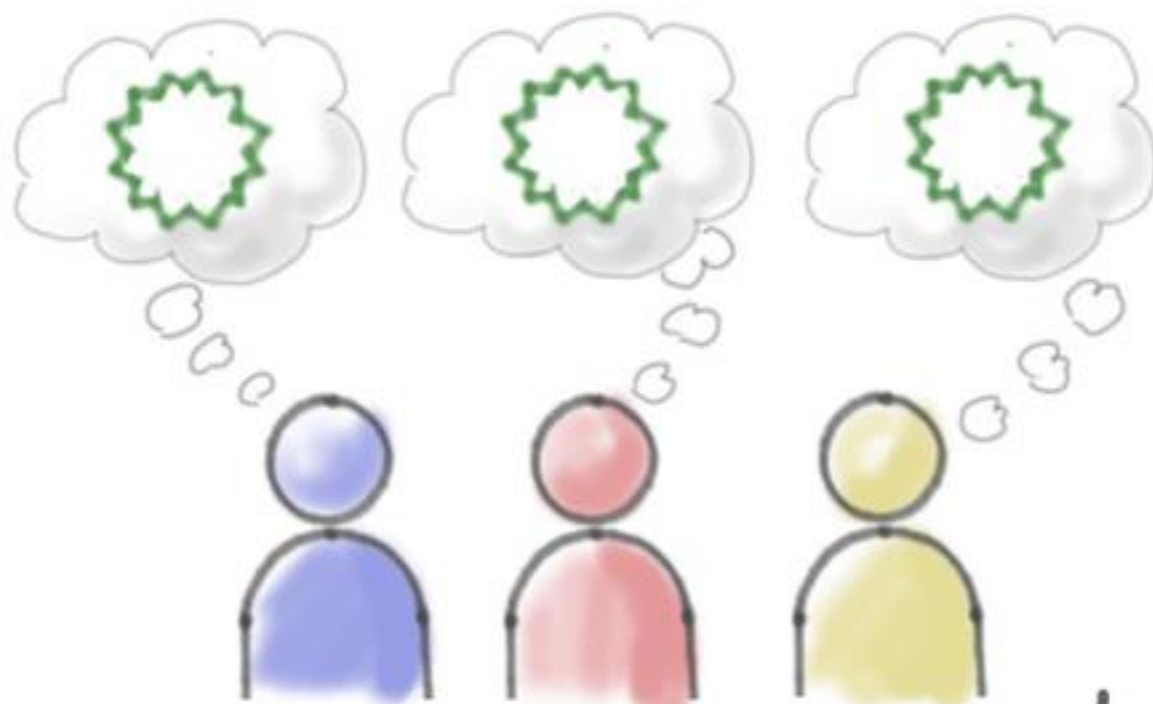


I'm glad we all agree.

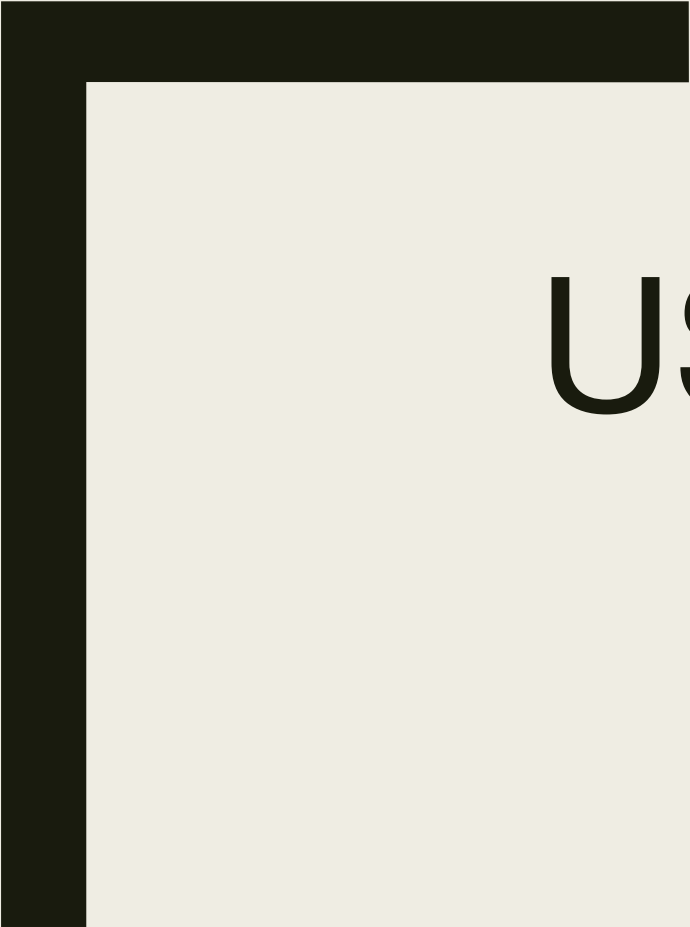




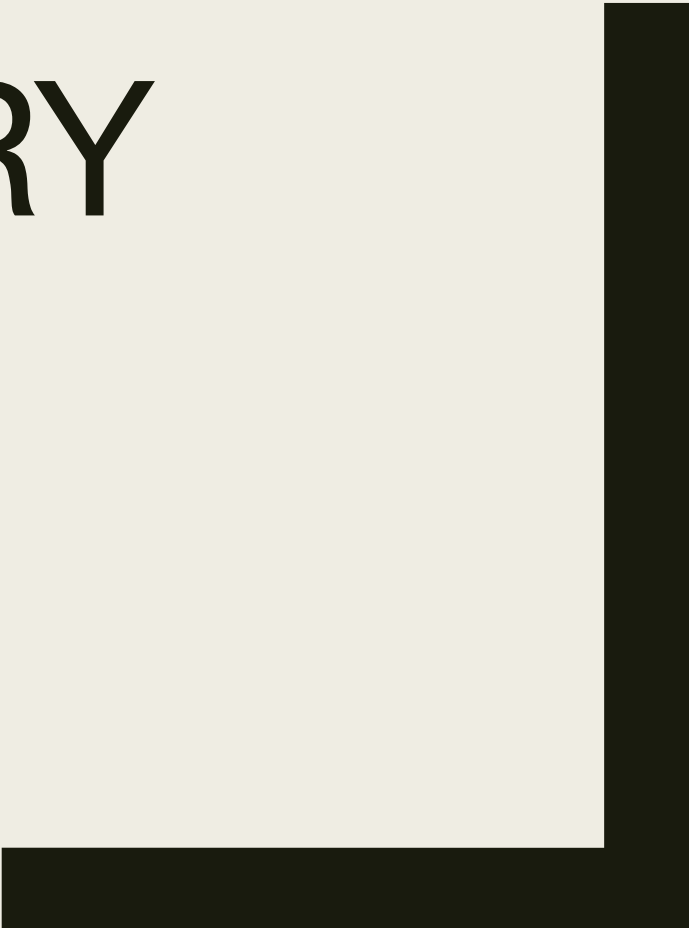
ah ha!



I'm glad we all agree!



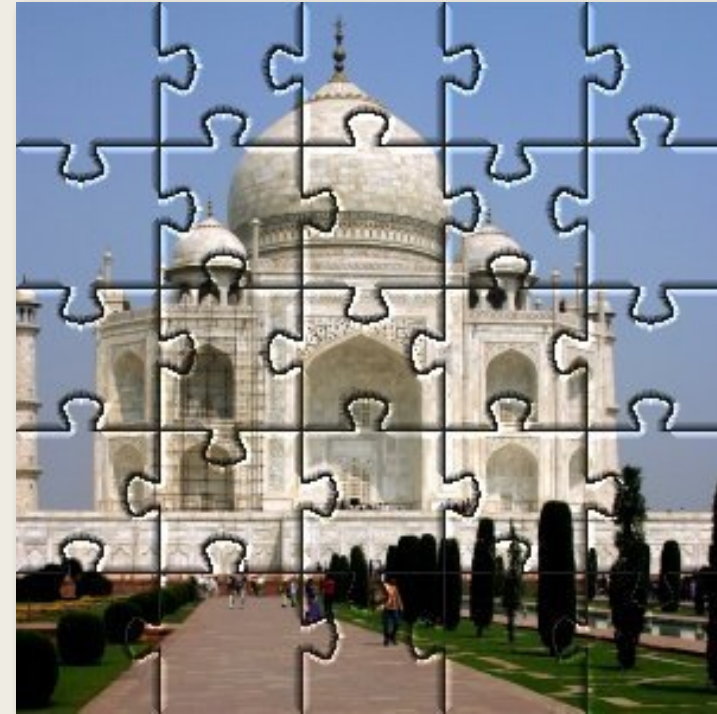
USER STORY MAPPING



Motivation



Backlogs are usually flat



How do you get the big picture?

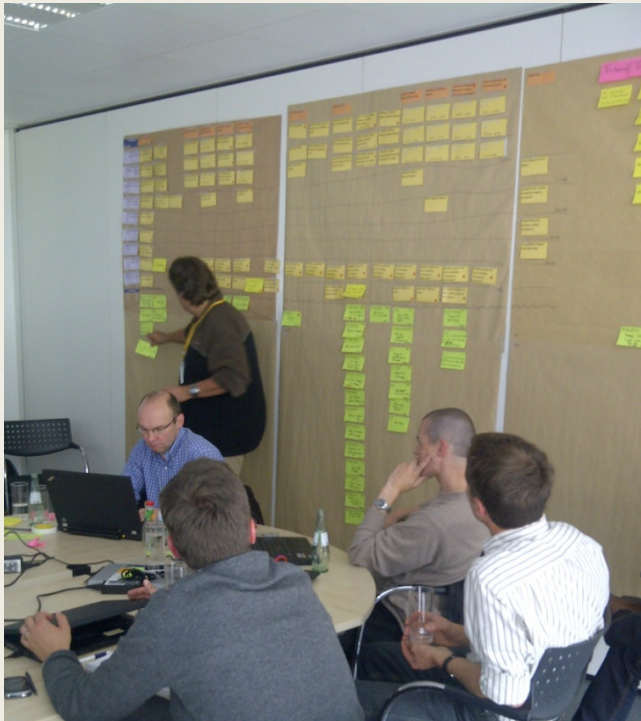
Source: Jeff Patton www.AgileProductDesign.com

Teilnehmer

Das ganze Team sollte am User Story Mapping beteiligt werden

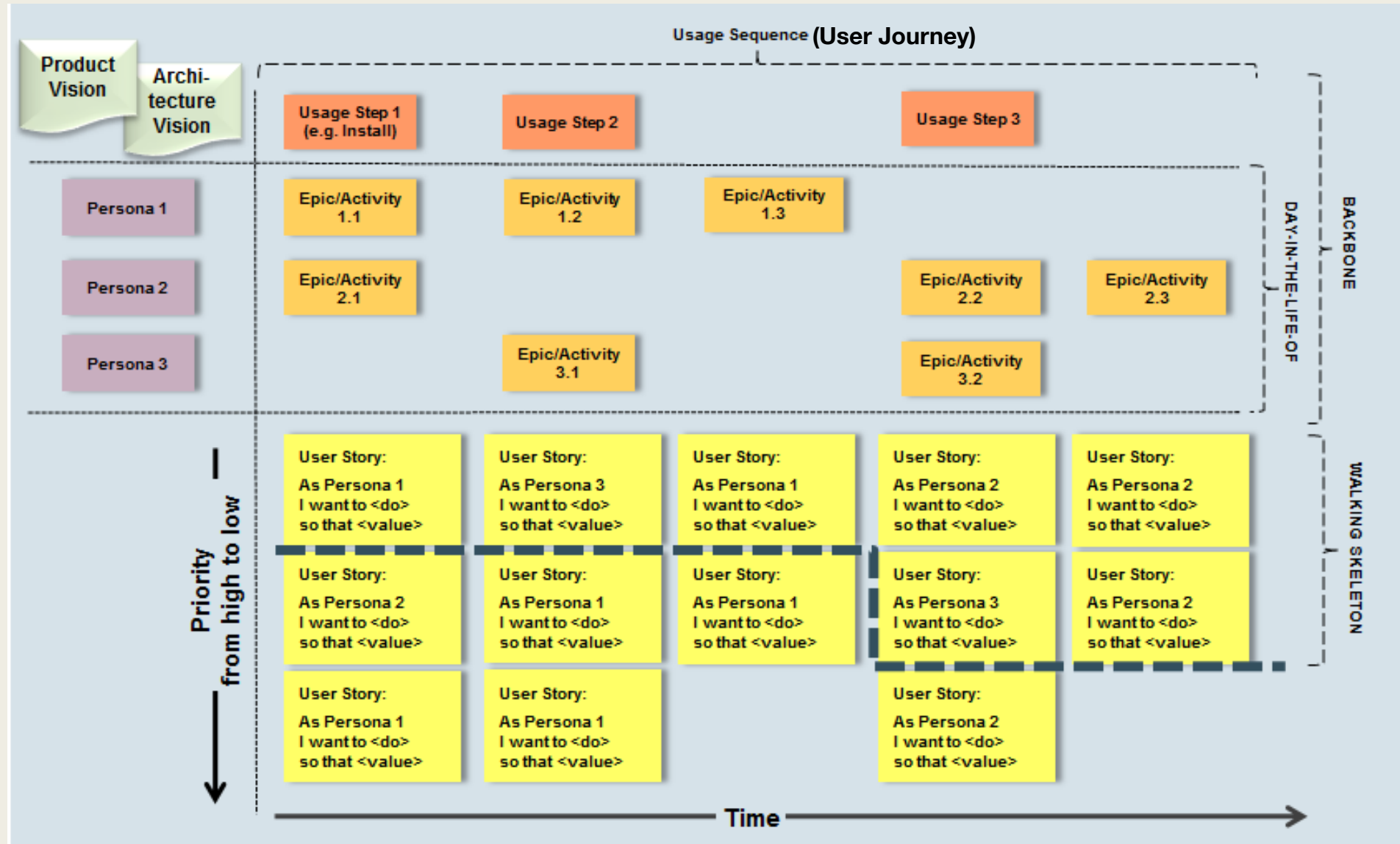
Erweitern und Validieren der User Story Map mit Kunden, End-Benutzern und anderen Projektbeteiligten





User Story Mapping in action: facilitated workshop

Wie sieht eine User Story Map aus?



User Story Mapping - Beispiel Web Shop

<https://manifesto.co.uk/user-story-mapping/>

**Product
search**

Product page

Checkout

Sort

Reviews

**Pay with
PayPal**

Filter

**Media
gallery**

**Pay with
credit card**

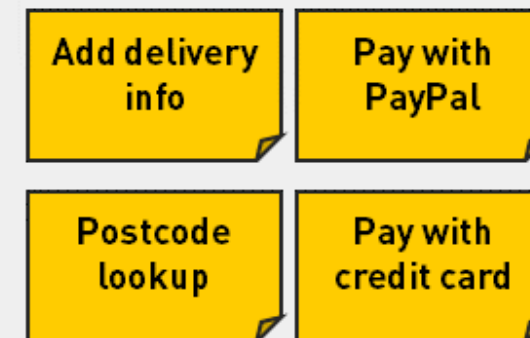
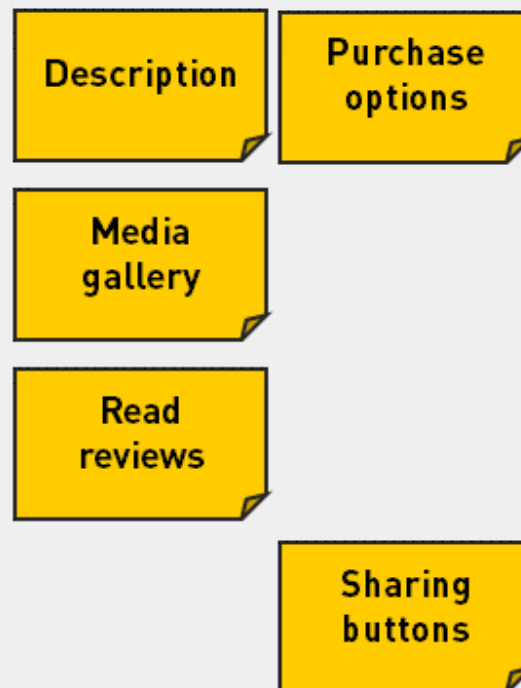
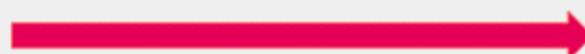
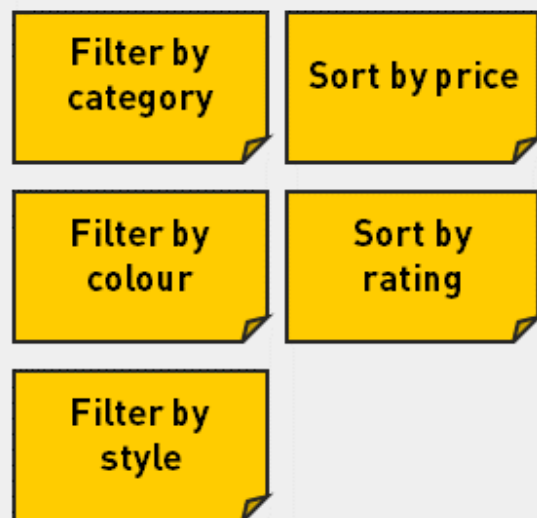
Description

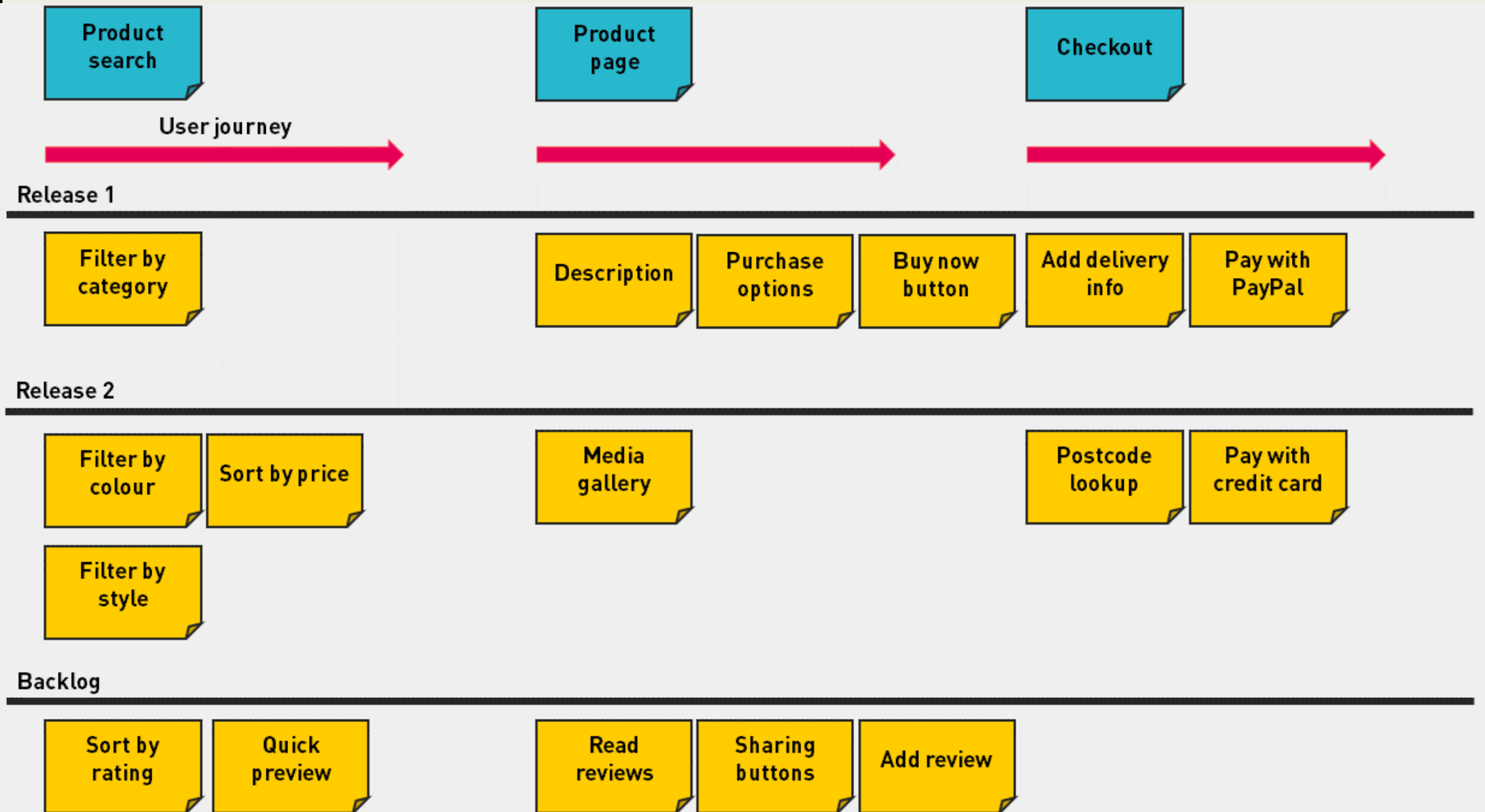
Product search

Product page

Checkout

User journey





User Story Mapping - Beispiel eMail

<http://winnipegagilist.blogspot.de/2012/03/how-to-create-user-story-map.html>



User Story Map – Beispiel SAP Transportation Management



Usage Dimension

- **Leading question:**

- *How will the customer work with your product?*

- **What you need to do:**

- *Identify the usage sequence for the product from a customer's perspective*

Purpose:

- *Ensure end-to-end coverage of the process, create a basis for more fine grained user stories*

Personas

Personas sind fiktive Charaktere, um die Bedürfnisse und Ziele von Benutzergruppen zu repräsentieren

- Personas stellen Benutzertypen dar und helfen, deren Verhalten, Ziele und Motivationen zu verstehen
- Personas stellen Benutzerbedürfnisse in den Mittelpunkt der Entwicklung
- Entwicklerteams können Empathie für die Benutzer entwickeln und bessere Entscheidungen über Funktionen und Design treffen
- Personas unterstützen die Kommunikation innerhalb des Teams und mit Stakeholdern, indem sie ein klares Bild der Zielbenutzer vermitteln

Personas (Beispiele) – SAP Transportation Management

Name	Carl Weber	Edi Sender	Steve Puller
Picture			
Role	Customer ordering via Web	Customer ordering via EDI	Dispatcher
Skills	Varies from little to enormous software know-how Could be John Doe	She is used to work with computers	Power user level very used to work with software
Goals	Easy, efficient and fast order entry Guidance for order entry	Smooth EDI process As little errors as possible for post processing	Enable efficient operations
Pain	Difficulty to understand system responses	Translation of EDI messages between industry standards	No sufficient overview on his assets
Points	Slow response time Error-prone system		

Personas

Context: Restaurant Table Reservation

Ravi Soni



AGE	58
EDUCATION	Ph.D, M.Tech
FAMILY	Wife, 2 children and 3 grandchildren
OCCUPATION	Senior Data Scientist
LOCATION	Amsterdam, Netherlands
TECH LITERTE	High

“ I have so many work responsibilities and want to spend some quality time with my family

Personality

Extrovert Analytical Reader
Thinker

Bio

Ravi is 58 who works in a organization with so many responsibilities. He has 34 years of experience in his field. He is senior data scientist and manages 3 teams in 2 different country. He lives with his wife and have 2 children. Both the children are married and one having 2 kids and another one is having 1 kid. He is busy with his work as he has to handle few teams. After all these responsibilities he still gives time to his family.

Goals

- Want to spend time with family in a dinner
- Find a nice restaurant with quality food and nice view
- Doesn't want to spend much time in table reservation process.

Frustrations

- So many restaurant options to choose from
- Large menu with a lot text without image
- Doesn't have much time to spend looking better options
- Long checkout process

Brands



Payment medium



Platform



User Dimension

- **Leading questions:**

- *Which types of users will work with your product?*
- *Which aspects of these personas are relevant for creating your product?*

- **What you need to do:**

- *Do research, e.g. user interviews, and derive the personas*

- **Purpose:**

- *With personas in mind it is easier to design for real users – personas facilitate empathy*

Building the Backbone

- **Leading question:**

- *What are the activities of each persona along the usage sequence of the product?*

- **What you need to do:**

- *Describe all the activities performed by one persona that make up that persona's "day-in-the-life-of"*
- *Focus on how they'll be using your product in the long run, not just the first delivery*

- **Purpose:**

- *Build the backbone of your user story map and thus a guiding structure for your product backlog*

Building the backlog: user stories

- **Collect user stories:** *As <persona> I want to <need> so that <value>*
 - Ensure the right granularity
- **Assess user stories**
 - Business value
 - Technical risk
 - Effort
- **Prioritize user stories**
 - Sufficient business value with feasible amount of effort and acceptable risk level
- **Draw the release line** – *you get a “Walking Skeleton”*
 - Ensure consistency – step back

User Stories Examples

As **Danny Driver**, I want to **block badly behaved passengers** so **they are never shown me again**.

As **Paula Passenger**, I want to **link the credit card to my profile** so that **I can pay for a ride faster, easier and without cash**.

As **Danny Driver**, I want to **add photos of my car in my profile** so that **I can attract more users**.

As **Paula Passenger**, I want **several available drivers to be displayed** so that **I can choose the most suitable option for me**.

Prioritising the backlog

- **Leading question:**
 - *Which user stories should be included into the next release / sprint?*
- **What you need to do:**
- Prioritise & estimate the user stories. You can use various methods:
 - Voting: In scope / Out of scope
 - Planning poker
 - T-shirt size estimation: S, M, L, XL
- **Purpose:**
 - *Define the viable scope / minimum marketable features*

Gruppenarbeit - “Systemanalyse”

Erstellt im Team (5-6 Teilnehmer je Gruppe) eine User Story Map (Personas, User Stories, User Story Map, Prototyp der Benutzeroberfläche)

Thema:

Online-

**Reservierungssystem für
Kino-Tickets**



Gruppenarbeit – Referate “Entwurf”

1. Version Control (git), Trunk based development vs Feature Branches; Continuous Integration, Continuous Delivery und Continuous Deployment (CI/CD)
2. Unit-Tests und Test Driven Development (Unit Tests, TDD, BDD, Code coverage, xUnit Test Patterns); Refactoring, Code Smells
3. Design Prinzipien (siehe <http://principles-wiki.net/>) – mind. 3 auswählen (zB SRP, DRY, IH/E, LoD, DIP); Design Patterns (siehe <https://refactoring.guru/design-patterns/catalog>) - bitte je nur 1 creational, structural, behavioural pattern auswählen (Composite, Façade, Strategy, Factory Method, Singleton (Anti-Pattern!), Observer); wissenschaftlicher Background zu Mustern im SE (Gang-of-four), Übersicht über andere Design Patterns, aktuelle Design Patterns (in der KI, Messaging Patterns)
4. Monolithische und Microservices Architekturen